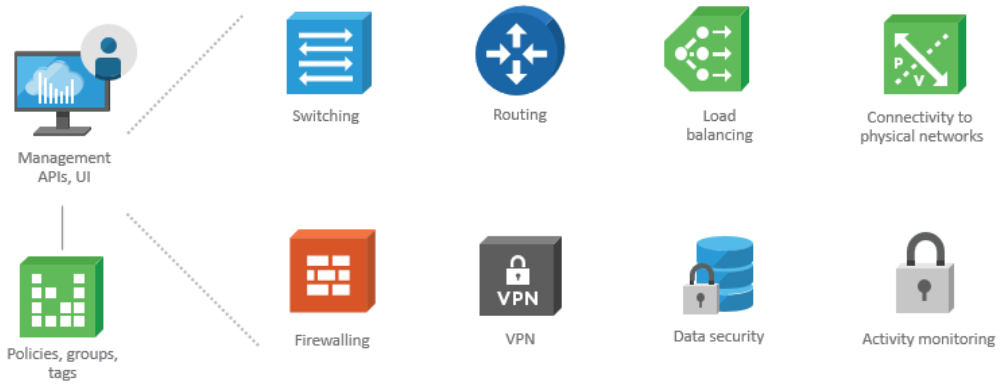
NSX Architecture Components Review

1. VMware NSX is the network virtualization and security platform for the Software-Defined Data Center. NSX enables the creation of entire networks in software and embeds them in the hypervisor layer, abstracted from the underlying physical hardware. All network components can be provisioned in minutes, without the need to modify the application. Micro-segmentation and granular security are delivered to the individual workload.

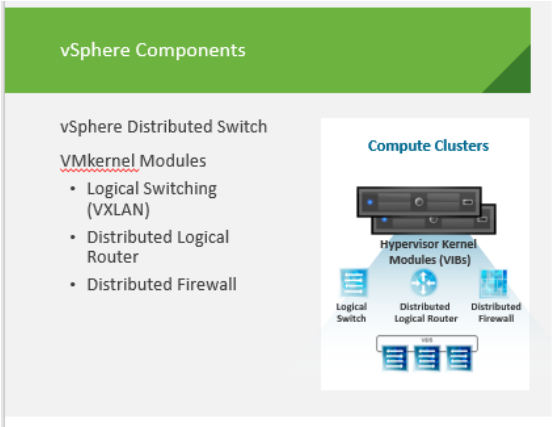
2. NSX provides more than just Intelligent Switching. NSX provides Network Virtualization and Security (In-Kernel Switching, Routing, and Firewalling, as well as Load-Balancing, Layer-2 Bridging, VPN, and more).

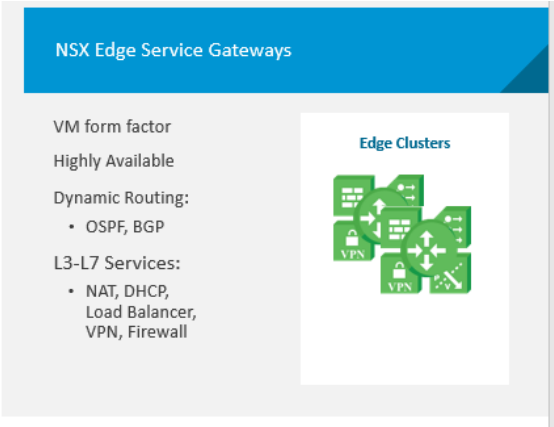


3. All NSX Functions are exposed through APIs. The APIs can be used by any Consumption tool or platform (vRealize Orchestrator, PowerNSX, vRealize Automation, OpenStack, vCloud Director, etc.) to configure NSX, as well as all day 2 operations (for creating and managing NSX objects).

4. Broad adoption of NSX impacts the Network Engineering team, just as the adoption of the Hypervisor impacted the System Administrators.

5. The NSX data plane is comprised of NSX modules running in the kernel of prepared ESXi hosts (VXLAN, DLR, DFW) and NSX Edge Service Gateways. Edge Service Gateways can modulate N/S flows, perform Dynamic Routing, Static Routing, and L3 – L7 Services (NAT, DHCP, VPN, Firewall, Load Balancer).

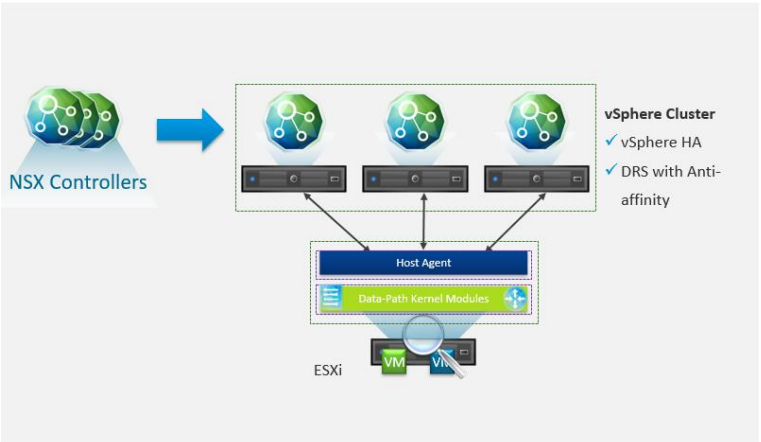




6. NSX Controllers provide Control Plane functions for Logical Switching and Logical Routing, support for ARP Suppression, and the elimination of Multicast for VXLAN support if needed.

7. The NSX control plane manages all logical networks and forwarding decisions. The control plane is separated from the data plane. Loss of the control plane does not impact the live applications.

8. NSX Controller virtual machines are deployed as part of the initial NSX configuration. These VMs are deployed with 4CPUs and 4GBs of RAM. Modifying their hardware specification is not supported. They should be deployed in odd numbers. By VMware's stated recommendation, the maximum number of controller nodes to deploy in a production environment is three.



Properties

- Virtual Form Factor (4 vCPU, 4GB)
- Data plane programming
- Control plane Isolation

Benefits

- Scale Out
- High Availability
- VXLAN - no Multicast
- ARP Suppression

9. Deploying three controllers provides High Availability for the control plane in the event of a controller failure. It also ensures workload distribution, as the workload of the control plane is

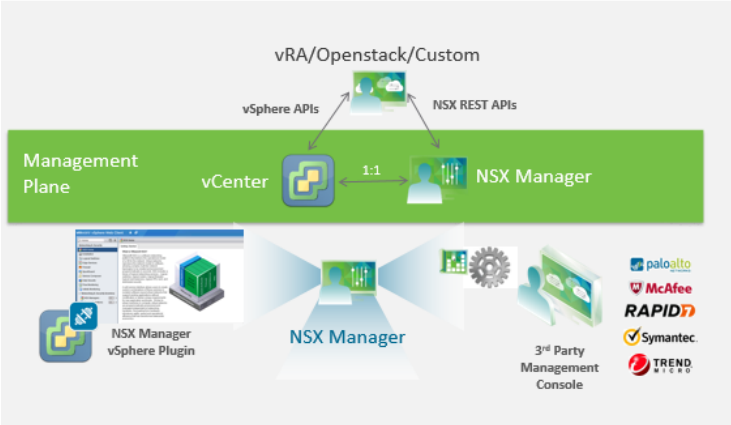
carried out by all controllers (all controllers are active).

10. NSX Manager is deployed from an OVF and runs as a virtual machine. It provides the Web Client plugin to vCenter and hosts the NSX REST APIs. Together, NSX Manager and vCenter provide the management plane for NSX.

11. A single NSX Manager can only be mapped to a single vCenter.

12. Switching, routing, and firewalling are implemented in the hypervisor of each ESXi host.

13. NSX Manager stores and deploys the OVFs for the NSX controllers, NSX Edges, and Control VMs.

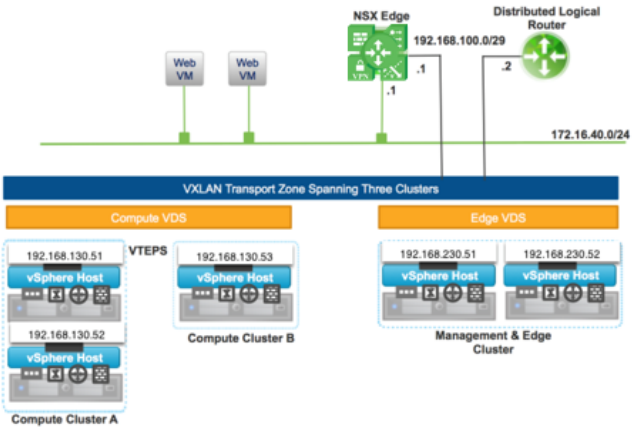


Runs as a Virtual Machine

Provisioning and Management of Network and Network services

- VXLAN Preparation
- Logical Network Consumption
- Network Services Configuration
- 1:1 mapping between an NSX Manager and vCenter Server
- Management UI and API for NSX
- vSphere Web Client Plugin
- Deploys NSX Controller and NSX Edge Virtual Appliances (OVF)

14. A transport zone defines the scope (boundary) of a logical switch. If a logical switch is created in a transport zone with one cluster mapped to a dedicated DVS, the logical switch will only be created on hosts in that cluster. Likewise, if a logical switch was created in a transport zone with two or more clusters, each with a dedicated DVS or a shared DVS, then the logical switch will be created across all hosts in those clusters.



A transport zone controls which hosts a logical switch can span

Dictates which clusters can participate in the use of a particular network

A logical switch can belong to only one transport zone

15. NSX provides two types of routing: distributed and centralized routing. Distributed routing is optimized for E/W routing and is carried out in the ESXi Kernel. Centralized routing is optimized for N/S routing and is carried out by NSX Edge(s).

16. In addition to N/S routing, NSX Edge(s)offer the following L3 – L7 services:
    a. DHCP
    b. DHCP relay
    c. VPN (L2VPN, SSL VPN, IPSec)
    d. Load Balancing
    e. NAT (Source & Destination)
    f. Firewalling

17. NSX Edge(s) can be deployed with support for scale-out and redundancy by enabling ECMP and High Availability.

18. VMware provides virtualized abstractions of CPU, Memory, Storage, Networking, and Security.

19. The DFW kernel modules communicate directly with the NSX Manager via the vsfwd service daemon.

20. NetCPA is an SSL client that communicates with the NSX Controllers on the control plane from the ESXi host.